One-to-Many Relationship (1:N)

Topic: 1:N & N:1 Cardinality   Exam Importance: High ⭐⭐⭐⭐

1. What is a One-to-Many Relationship?

A **One-to-Many (1:N) relationship** means that one entity from entity set A can be associated with multiple entities from entity set B, while each entity in set B is associated with at most one entity in set A.

```
CUSTOMER (1)   ----- < PLACES > -----   ORDER (N)
```

- **One customer** can place many orders.
- **One particular order** belongs to only one customer.

Mapping: C1 → O1, C1 → O2, C1 → O3, C2 → O4, C2 → O5. Customer C1 appears multiple times because one customer can place multiple orders.

2. Types of Cardinality

Cardinality Ratio	Meaning
1:1	One-to-One
1:N	One-to-Many
N:1	Many-to-One
M:N	Many-to-Many

> **Exam Tip:** Always write Many-to-Many as **M:N**, not *M:M*, because the number of entities on each side need not be equal.

3. Converting 1:N ER Model into Relational Model ⭐⭐⭐⭐

Initially, we define relational tables for each entity set:

Customer Table

Customer_ID PK	Name	City
C1	A	Jalandhar
C2	B	Delhi
C3	C	Mumbai
C4	A	Mumbai

Order Table

Order_ID PK	Item	Cost
O1	Bucket	₹1000
O2	Shoes	₹2000
O3	Shirt	₹1500
O4	Jeans	₹2000

4. Relationship Table & Descriptive Attributes ⭐⭐⭐

A relationship itself can have descriptive attributes (e.g., `Date` indicating when the order was placed). If initially represented as a separate table:

PLACES Table (Initial Unmerged State)

Customer_ID FK	Order_ID FK	Date (Descriptive Attribute)
C1	O1	01-09-2026
C1	O2	02-09-2026
C2	O3	03-09-2026
C2	O4	04-09-2026

5. Primary Key Determination for 1:N ⭐⭐⭐⭐

In the `PLACES` table:

- `Customer_ID` **repeats** (C1 appears for O1, O2). Therefore, it cannot uniquely identify rows and ❌ *cannot be the Primary Key*.
- `Order_ID` **never repeats** because each order is placed by exactly one customer. Therefore, `Order_ID` uniquely identifies rows and ✅ *must be the Primary Key*.

> 🔥 **Golden Rule for 1:N:** The Primary Key of the entity on the **MANY (N) side** becomes the Primary Key of the relationship table.

6. Schema Reduction: Merging Tables ⭐⭐⭐⭐

Initially, we have **3 tables** (Customer, Order, Places). Since `Order_ID` is the primary key of `PLACES`, we can merge `PLACES` directly into the `ORDER` table.

Why Can't We Merge Customer + Places?

If we merge `CUSTOMER` with `PLACES`, customer information (Name, City) would repeat for every order placed by that customer, creating severe data redundancy and anomalies.

Final 2 Merged Tables

1. Customer Table

Customer_ID PK	Name	City
C1	A	Jalandhar
C2	B	Delhi
C3	C	Mumbai
C4	A	Mumbai

2. Order Table (Merged with PLACES)

Order_ID PK	Item	Cost	Customer_ID FK	Date
O1	Bucket	1000	C1	01-09-2026
O2	Shoes	2000	C1	02-09-2026
O3	Shirt	1500	C2	03-09-2026
O4	Jeans	2000	C2	04-09-2026

7. Comparison: 1:1 vs. 1:N Relationships

Feature	One-to-One (1:1)	One-to-Many (1:N)
Duplicate First-Side Key in Rel Table	❌ No	Depends on cardinality side
Duplicate Many-Side Key in Rel Table	—	❌ No
Relationship Table PK	Either side's key	**Many-side key strictly**
Merged with Which Entity?	Either side	**MANY side strictly**
Final Table Count	2 tables	2 tables

🚨 GATE / UGC NET / BCA Exam Rules

1. **Rule 1:** For `A (1) —— N —— B`, the key of **B (MANY side)** becomes the Primary Key of the relationship.
2. **Rule 2:** The key of the ONE side can repeat in the relationship table.
3. **Rule 3:** The key of the MANY side can never repeat in the relationship table.
4. **Rule 4:** Descriptive attributes belong to the relationship and move with the merged table.
5. **Rule 5:** Always merge the relationship table with the **MANY-side entity table**, reducing 3 tables to 2.

🧠 Quick Revision & Memorization

Workflow: 1:N → Identify MANY side → Many-side Key = PK of Relationship → Merge with MANY side → 3 Tables become 2 Tables.

🔥 Key Memory Phrase: "In a One-to-Many relationship, find MANY first; its primary key becomes the primary key of the relationship table, and the relationship merges with the MANY side."